



Department of Computer Science
Course Title: Evolutionary Computing (2025)
Assignment 4: Feature selection with an adaptive multi-objective

1. Introduction

Feature selection removes irrelevant or redundant variables to produce a smaller, more informative feature set, which reduces overfitting, improves interpretability, and speeds up training. Xue et al. present **MOBGA-AOS**, a multi-objective binary genetic algorithm tailored to this task. Because genetic algorithms are population-based and approximate Pareto fronts well, they suit feature selection. The key innovation is adaptively choosing crossover operators during the search to improve performance.

2. Problem Statement

Feature selection is a bi-objective problem: reduce the number of features while keeping (or improving) classification accuracy. Traditional MOEAs like NSGA-II work, but their effectiveness depends heavily on which crossover operators are used, and no single operator dominates across datasets. **MOBGA-AOS** extends NSGA-II with adaptive operator selection: solutions use binary encoding and multiple crossover operators whose selection probabilities are continually updated based on how often they produce non-dominated offspring. This feedback-driven mechanism biases the search toward effective operators over time, making the algorithm more robust and efficient at tracing a high-quality Pareto front.

2.1 Representation

Each individual is represented as a binary array with length equal to the number of features in the dataset.

1 indicates the feature is selected, 0 not selected.

2.2 Fitness Evaluation

- **Objective 1:** Classification error achieved by evaluating solutions using k-NN ($k = 3$) with n fold cross validation ($n = 3$). Must be Minimized.

2.3 Operators

2.3.1 Crossovers

Uses a pool of 2 crossover operators: shuffle, reduced surrogate. Select crossover operator via roulette wheel based on OSP.

- **Shuffle crossover:** Randomly shuffle genes of parents, then apply single-point crossover on shuffled parents.
- **Reduced surrogate crossover:** Only allow crossover points where parents have different genes. Randomly select one such point and apply single-point crossover.

2.3.2 Mutation

- Uniform mutation with rate of P_m .
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2.3.3 Environmental Selection

- Use NSGA-II's original fast nondominated sorting and crowding distance to select survivors.
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2.4 Credit assignment & Updating of OSP

To find out about the process of assigning credits and updating OSP (Operator Selection Probability vector), read section 3.5 and section 3.6 in the original paper.

3.Tasks

- Implement MOBGA-AOS and apply it to the 6 provided datasets individually. To validate your results, you can compare your final obtained pareto front with those presented in the original paper. They should be at least relatively close.
- Plot the obtained pareto front of your implementation.
- Suggest a solution to converge to optimal solutions more quickly for datasets with a larger number of features (more than 250 features). Justify your solution.
- Plot the hypervolume value for each generation.
- report the average and standard deviation of the final values for IGD1 and HV (hypervolume) for each run(3 times)
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Submission Guidelines

Programming Languages: Python, MATLAB

Allowed Libraries: NumPy, Pandas, Matplotlib , You can use pre-defined libraries to compute IGD and HV values.

File Naming: Compress all files into a ZIP named: Lastname-StudentNumber.zip